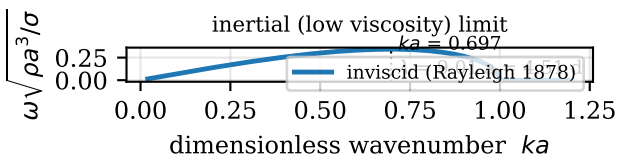
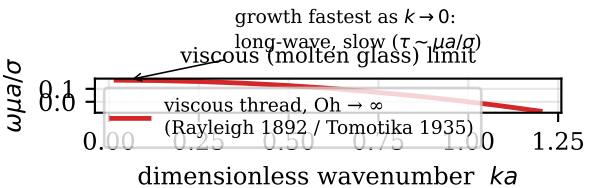

inertial (low viscosity) limit
$ka = 0.697$
inviscid (Rayleigh 1878)
$\omega\sqrt{\rho a^3/\sigma}$
0.25
0.00
0.00
0.25
0.50
0.75
1.00
1.25
dimensionless wavenumber ka


growth fastest as $k \to 0$:
long-wave, slow ($\tau \sim \mu a/\sigma$)
viscous (molten glass) limit
viscous thread, Oh → ∞
(Rayleigh 1892 / Tomotika 1935)
$\omega\mu a/\sigma$
0.1
0.0
0.00
0.25
0.50
0.75
1.00
1.25
dimensionless wavenumber ka